



André F. Pinheiro Aleixo

FPGA & Embedded Systems Engineer



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San Sebastián, Basque Country, Spain

Professional Summary

Ever since I was a child, I have loved opening electronics, tracing how they work, and fixing what is broken. That habit never left me; it shaped how I think about hardware, firmware, and systems. What drives me is a simple mission: to learn as much as possible, as deeply as possible, and to turn that curiosity into reliable, useful engineering.

Education

University of Deusto

Bilbao, Spain

Master's Degree in Automation, Electronics and Industrial Control

2019 – 2020

- Master's Degree in Automation, Electronics and Industrial Control

Skills: LabVIEW, Verilog, Siemens Tia Portal, Kuka Robots, TI DSPs, Device Drivers, C#, NI LabVIEW, Yocto Project, MATLAB

Universidade Federal de Pernambuco

Recife, Pernambuco, Brazil

Bachelor's Degree in Electrical and Electronics Engineering

2013 – 2018

- Bachelor's Degree in Electrical and Electronics Engineering

Skills: Field-Programmable Gate Arrays (FPGA), Verilog, ChibiOS, FPGA, Electronics Design, Intel Quartus Prime, Bare-Metal Embedded, Proteus, C++, MATLAB, Real-Time Operating Systems (RTOS), C

Professional Experience

INZU Group (Ikergune)

San Sebastián, Basque Country, Spain

Electronics Department Lead

January 2023 – Present

- Development of vision systems using NIR and SWIR spectral cameras for process analysis, creating custom C# software for real-time image acquisition and processing with support for RTSP and HTTP streaming protocols
- Designed and implemented FPGA IP cores for convolutional kernel acceleration on Xilinx UltraScale+ devices, developing high-throughput data pipelines for real-time image processing

- Integrated FPGA acceleration into embedded edge computing platforms running Yocto Linux, developing C++ drivers and middleware to interface with the FPGA and manage data flow
- Engineered industrial communication solutions integrating IO-Link and HTTP protocols with machine networks, implementing protocol stacks for robust data exchange and control
- Developed custom sensor systems using Infineon microcontrollers running FreeRTOS, including PID control loops and SPI peripheral communication, with PCB design handled via Altium Designer, for particulate matter measurement
- Designed and implemented stepper motor control systems using XMC microcontrollers integrated with FreeRTOS, developing trajectory control algorithms including acceleration and deceleration profiles in C/C++ for precise motion management

Skills: Yocto Project, Verilog, FPGA, C++, Embedded C, HDL Designer, FPGA prototyping, Bash, VHDL, Field-Programmable Gate Arrays (FPGA), C, CMake, Bare-Metal Embedded, Device Drivers, Data Acquisition, Embedded Linux, Embedded Systems, Hardware Description Language, Firmware, I2C, IO-Link, Kernel, Devicetree, U-Boot, BOOT, Bootloader, OpenCV, Computer Vision, Machine Learning

INZU Group (Ikergune)

San Sebastián, Basque Country, Spain

Software Engineer

August 2022 – January 2023

- Designed and implemented the full integration pipeline between a custom CAD/CAM system and a robotic simulation API, using C++ to automate simulation, path planning, and KUKA code generation
- Developed both the API interface and post-processor, handling the complete workflow from trajectory acquisition to robot-compatible output
- Adapted toolpaths from upstream CAM software to KUKA syntax, incorporating kinematic validation, axis limits, and machine-specific parameters
- Validated generated programs through simulation and deployment in the robotic cell, ensuring precise and collision-free execution

Skills: C++, CMake, Application Programming Interfaces (API), Automation, Industrial Robots, Robot Framework, CAD/CAM Software

INZU Group (Ikergune)

San Sebastián, Basque Country, Spain

Robotics Integration Engineer

May 2021 – August 2022

- Developed and configured the complete KUKA robot control system using WorkVisual, including programming of robot motion sequences and system diagnostics
- Engineered safety systems for the robotic cell, implementing both robot-integrated safety functions and safety zone monitoring using PROFIsafe, the safety protocol stack over PROFINET, ensuring compliance with industrial safety standards (e.g., SIL/PL)
- Performed precision and repeatability verification using laser tracker systems, developing custom C# scripts to automate measurement sequences, data acquisition, and error analysis

Skills: Robot Framework, Protocol Stacks, PROFINET, PROFIsafe, IO-Link, EtherCAT, C++, C#

INZU Group (Ikergune)
Automation & Software Developer

Elgoibar, Basque Country, Spain
September 2020 – May 2021

- Developed Python scripts to automate and optimize internal industrial processes, improving efficiency and repeatability in manufacturing workflows
- Built C# applications to integrate user interfaces and machine controls, streamlining human-machine interaction and data flow
- Participated in computer vision projects, contributing to the development of inspection and monitoring systems using cameras and image processing techniques
- Contributed to the design and implementation of a patented industrial control system for automated powder feeding equipment, integrating motor control, pressure regulation, and sensor feedback loops
- Developed control logic using PLCs and integrated SCADA systems for real-time process monitoring, data acquisition, and system supervision

Skills: SCADA, PLC Siemens, PLC Programming, Programmable Logic Controller (PLC), SIMATIC STEP 7, PyTorch, Tkinter, Python (Programming Language), Python, Anaconda, Windows Presentation Foundation (WPF), WPF Development, C#

INZU Group (Aingura IIoT)
System Engineer

Bilbao, Basque Country, Spain
February 2020 – September 2020

- Developed embedded C code for a Texas Instruments DSP to acquire digital sensor signals and communicate with a C# application via SCI (Serial Communication Interface)
- Built a C# application to acquire data from an industrial KUKA robot over TCP/IP (via KUKAVARPROXY) and interface with the DSP for synchronized data handling and storage
- Created LabVIEW RT modules on CompactRIO for real-time acquisition of digital signals and robot position data via TCP/IP, including data management and logging
- Integrated external sensors to capture physical signals related to robot movement, ensuring signal consistency across different acquisition systems
- Performed system calibration, trajectory setup, and signal comparison to validate synchronization and analyze timing behavior across hardware and software layers

Skills: Embedded Systems, Signal Processing, Internet Protocol Suite (TCP/IP), C#, LabVIEW, Kuka Robot, Data Analysis, MATLAB, NI LabVIEW, TI DSPs, Kuka Robots, Industrial Robots, Embedded C, Object-Oriented Programming (OOP)

Asa Branca Rocket Design —
Academic Project
Lead of Telemetry Team

Recife, Pernambuco, Brazil
March 2018 – July 2019

- Helped develop the telemetry subsystem for rockets, drones, and cubesats, focusing on real-time sensor data acquisition, processing, and inter-subsystem communication
- Programmed firmware in C using ChibiOS/RT on STM32 boards, ensuring a scalable and deterministic embedded architecture for critical flight operations

- Integrated sensors via SPI, I²C, and UART, optimizing driver-level communication and RTOS task management

Skills: Embedded Systems, Linux, Real-Time Operating Systems (RTOS), C (Programming Language), Task Management, Data Acquisition, Satellite Systems Engineering, ChibiOS, Universal Asynchronous Receiver/Transmitter (UART), STM32, SPI, I²C, Spacecraft Telemetry

Universidade Federal de Pernambuco

Recife, Pernambuco, Brazil

Researcher

July 2016 – January 2019

- Developed analog and embedded systems for biomedical therapy and monitoring applications
- Developed an electrical stimulator for gait improvement in Parkinson's patients, using MSP430 microcontroller and real-time inertial feedback (I²C accelerometers/gyroscopes) for stimulation timing control
- Designed a power control and switching circuit for hyperhidrosis treatment, using CMOS transistor arrays, MSP430-based PWM modulation, LTspice simulations, and PCB prototyping in Proteus
- Contributed to experimental validation and interface control of FES systems for neuromuscular stimulation

Skills: System on a Chip (SoC), C (Programming Language), Biomedical Electronics, Electronics Design, Embedded Systems, MATLAB, Signal Processing, MSP430, Launchpad TI (Texas Instruments), Bare-Metal Embedded, Real-Time Operating Systems (RTOS), I²C

Universidade Federal de Pernambuco

Recife, Pernambuco, Brazil

Undergraduate Researcher — NFC Vital

July 2017 – July 2018

Sign Acquisition System

- Developed a passive embedded system based on the NXP/Freescale FRDM-K64F microcontroller, powered via NFC and capable of acquiring vital signs such as heart rate and blood oxygen saturation without the need for an external power source
- Designed and integrated an NFC-based energy harvesting architecture using the NTAG I²C Plus interface (Class 6), enabling biomedical signal acquisition (MAX30100) and seamless wireless communication with an Android application
- Project presented at CBEB (Brazilian Congress of Biomedical Engineering), the largest biomedical engineering conference in Brazil, and awarded the Ricardo Ferreira Award to the Talented Young Scientist, recognized as the best undergraduate scientific research of the year (2018)

Skills: System on a Chip (SoC), Android Studio, Embedded Systems, NFC, NXP, Signal Processing, Radio-Frequency Identification (RFID), Bare-Metal Embedded, Biomedical Electronics, Freescale

CETENE

Recife, Pernambuco, Brazil

Internship — FPGA Engineer

August 2015 – December 2015

- Contributed to the initial development and simulation of a custom SDRAM controller IP for Altera FPGAs, using VHDL and ModelSim testbenches
- Contributed to the development and simulation of FPGA components for a low-power RFID reader project, aligned with the EPCglobal Class 1 Gen 2 protocol

- Assisted in the integration of Alien RFID Readers into a plant inspection application, supporting configuration, testing, and operational validation

Skills: Radio Frequency (RF), FPGA prototyping, VHDL, RFID Applications, Intel Quartus Prime, ModelSim, SystemVerilog

Publications

Wearable Device for Acquisition of SpO₂ and Heart Rate

André F. P. Aleixo; Euller G. Lima; Érico C. Leite; Ana V. M. Inocêncio; Lucas T. Lins; Marco A. B. Rodrigues 2019

- *Book chapter* - DOI: 10.1007/978-981-13-2119-1_90

Functional Electrical Stimulator for Treatment of Patients with Foot Drop

Jamile T. D. Alves; Marco A. B. Rodrigues; André F. P. Aleixo 2019

- *Book chapter* - DOI: 10.1007/978-981-13-2119-1_33

Virtual Reality Game Development Using Accelerometers for Post-stroke Rehabilitation

Gustavo R. P. Esteves; Bruno A. M. Miranda; André F. P. Aleixo; Malki-Shedheq B. C. Silva; Marco A. B. Rodrigues 2019

- *Book chapter* - DOI: 10.1007/978-981-13-2119-1_89

Languages

- **English:** Professional working proficiency
- **Portuguese:** Native or bilingual proficiency
- **Spanish:** Full professional proficiency

Technical Certifications

- **Verilog for an FPGA Engineer with Xilinx Vivado Design Suite**
- **Embedded Linux Using Yocto**
- **Programación TPE, Nivel B — FANUC Europe**
- **Certified LabVIEW Associate Developer** (National Instruments)